Master's Programme in Statistics and Machine Learning

732A99 Machine Learning

Computer Lab 1 Block 1

Lab Report

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1 Assignment 1

2 Assignment 2

3 Assignment 3 - Logistic regression and Basis Function Expansion

Visualization of the data shows that there are values equal to zero in the measures, including plasma glucose which is one of the predictors of interest in this assignment. `Glucose = 0` is not a feasible value, but rather representing a missing value. All cases (n=5) with plasma glucose equal to zero were excluded from the data before the regression analysis.

3.1 Scatterplot diabetes vs age

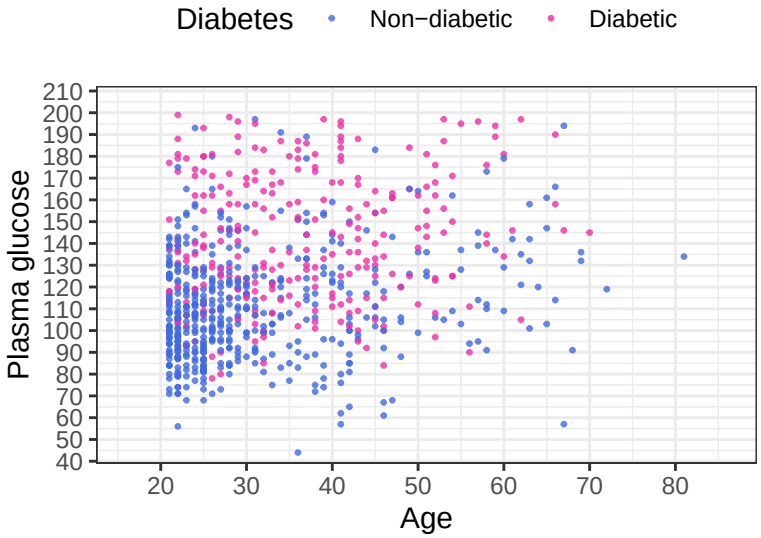


Figure 1: Plasma glucose levels in relation to age. Blue dots represent non-diabetic subjects, the pink dots diabetic subjects.

Plotting plasma glucose levels vs. age shows that the two response groups, *Non-diabetic* and *Diabetic*, show a substantial overlap. `Diabetes` would therefore not be easily predicted by *Age* and *Glucose* in a standard logistic regression model which would not be able to derive a straight line clearly separating the groups.

3.2 Logistic regression model

The vector form of the probailistic equation for a logistic regression model is

$$P(Y = 1 \mid x) = \text{sigmoid}(\theta^T x) \tag{1}$$

where the sigmoid function is

$$sigmoid(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}} \tag{2}$$

For *Diabetes* as the binary target/response (y), and *Glucose* and *Age* as features/predictors (x1, x2):

$$P(Y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 \cdot x_1 + \theta_2 \cdot x_2)}} \tag{3}$$

Training a logistic regression model on the (entire) provided data set, using glm() in R, results in the coefficients $\theta_0 = -6.2252$, $\theta_1 = 0.0384$, and $\theta_2 = 0.0231$, all with significant p values, and the corresponding probabilistic equation is then

$$P(Y = 1 \mid x_1, x_2) = \frac{1}{1 + e^{-(-6.2252 + 0.0384x_1 + 0.0231x_2)}} \tag{4}$$

The training misclassification error for this model (with r = 0.5) is 0.26, i.e approximately one out of four observations are misclassified.

3.3 Scatterplot colored by predicions

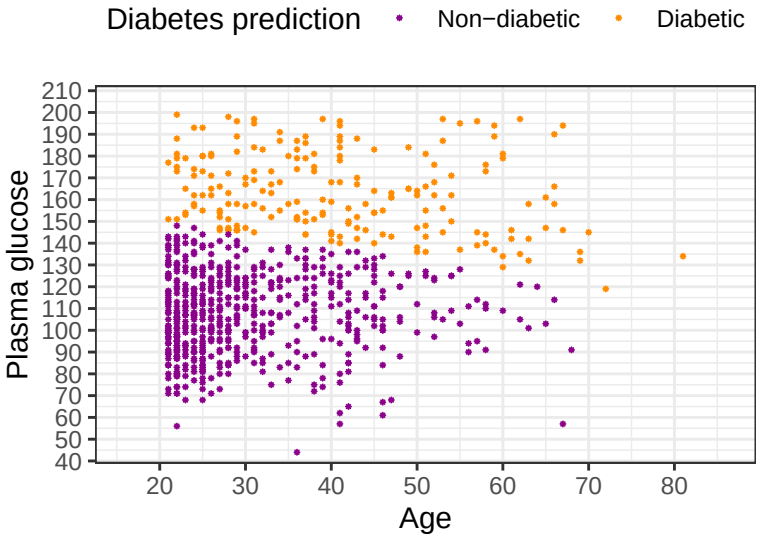


Figure 2: Plasma glucose levels in relation to age. Purple stars represent subjects predicted to be non-diabetic, the orange stars subjects predicted to be diabetic.

Comparing the predicted diabetes classifications to the actual diabetes status as shown in Figure 1 it looks like there will be some misclassifications, possibly more so for subjects in the lower, middle part of the scatter, where diabetic an non-diabetic subjects are overlapping, and for the non-diabetic elderly subjects.

3.4 Decision boundary equation

Setting the probabilistic equation in Equation 4 equal to `r` (here `= 0.5`) results in this approximate decision boundary equation, with the fitted coefficients and $x1 = Glucose$ and $x2 = Age$:

$$-6.2252 + 0.0384 \cdot Glucose + 0.0231 \cdot Age = 0 \tag{5}$$

Solving this for *Glucose* gives the approximate linear decision boundary line

$$Glucose = 161.97 - 0.6 \cdot Age \tag{6}$$

3.5 Scatterplot with decision boundary line for r = 0.5

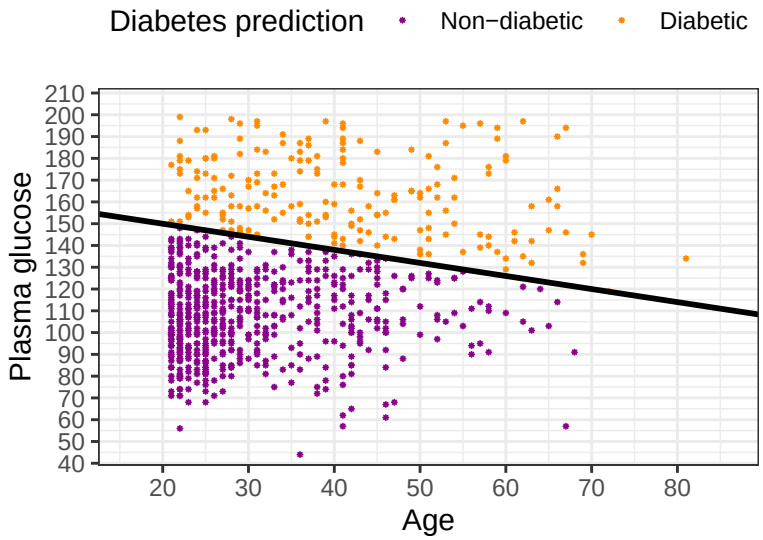


Figure 3: Predicted diabetes classifications with threshold r = 0.5. The added black line is the decision boundary line.

The boundary line means that there will likely be misclassifications both ways in the mixed "middle" group. The elderly non-diabetics will risk being misclassified as diabetic according to the negative slope of the linear boundary imposing increasingly harder restrictions on acceptable plasma glucose level for a non-diabetic subject.

3.6 Scatterplot with decision boundary line for r = 0.2 and r = 0.8

The equation for the linear decision boundary line for two predictors/features can generically be expressed as

$$x_1 = \frac{\log\left(\frac{r}{1-r}\right) - \theta_0}{\theta_1} - \frac{\theta_2}{\theta_1} x_2 \tag{7}$$

which means the intercept changes with r, the slope does not.

With r = 0.2 are more individuals predicted to be diabetic, with r increased to 0.8 instead fewer individuals. The training misclassification errors are 0.36 and 0.31 resepectively for the 0.2 and 0.8 cases, both higher than for the 0.5 cutoff case.

3.7 Basis function expansion

Fitting a new logistic regression model after adding new features

$$z_1 = x_1^4, \quad z_2 = x_1^3 x_2, \quad z_3 = x_1^2 x_2^2, \quad z_4 = x_1 x_2^3, \quad z_5 = x_2^4$$

results in a much more complex model but a training misclassification error (with r = 0.5) of 0.24 which is slightly better than the linear model. The decision boundary is now not linear but curvilinear, changing the appearance of the Glucose vs Age scatterplot (see Figure 5). The most obvious difference from the previous model is that older subjects are "allowed" to have higher plasma

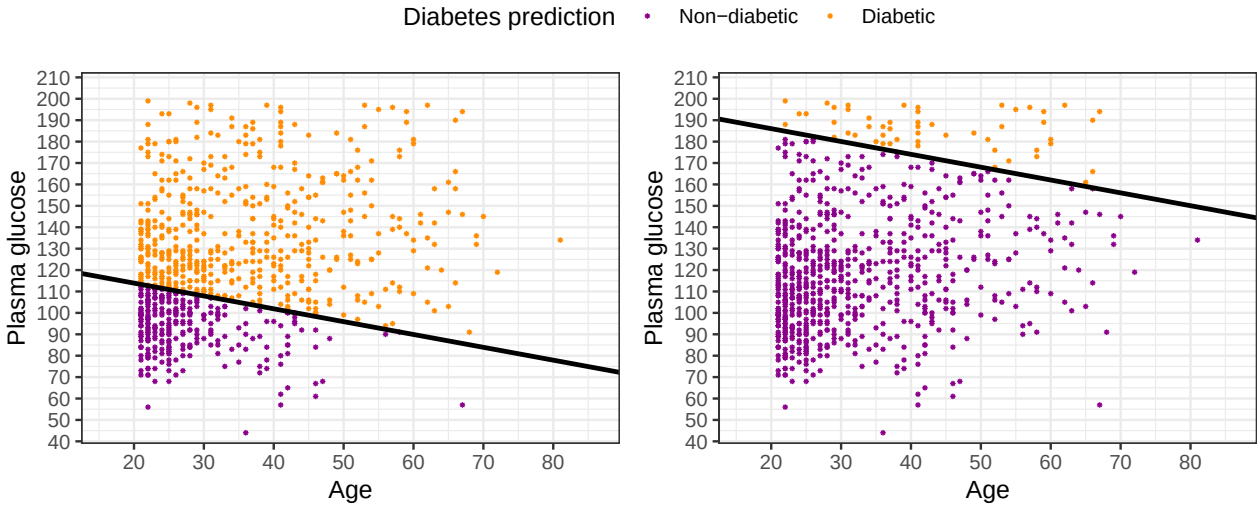


Figure 4: Predicted diabetes classifications for thresholds r = 0.2 (left) and r = 0.8 (right). Black lines indicate the decision boundaries.

glucose levels without being predicted to be diabetic, which is a pattern that can be discerned also in the actual data.

The *theta3* to *theta7* coefficients (for the *z1* to *z5* terms) are very small (magnitude $10^{-7} or less), and individually non-significant (p values above 0.6), but they are multiplied with numerically high values for *Glucose* (raised to up to 4) in the polynomial, and the AIC reduction suggests an improved overall fit for the model.

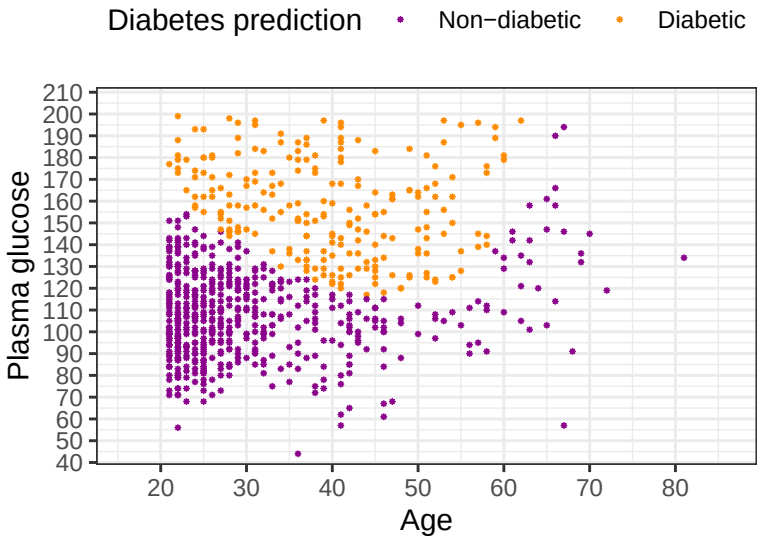


Figure 5: Predicted diabetes classifications for the extended logistic regression model with polynomial basis function expansion.

4 Theory

A R code

A.1 Assignment 1

A.2 Assignment 2

A.3 Assignment 3

A.3.1 Assignment 3 — Functions

```
################################################################################
# function definitions

################################################################################
#' A3_classifications_and_plots
#' Adds predictions based on provided predicted probabilities
#' and decision threshold, and creates a plot colored by diabetes prediction
#' and a plot colored by the predicion being correct or not,
#' the latter is extra so not presented in report.
#'
#' @param data_pred_prob A tibble with data and predicted probabilities
#' @param r Classification threshold
#' @param glucose_range Numerical vector with min and max of Glucose values
#' @param age_range Numerical vector with min and max of Age values
#'
#' @returns A list of:
#' data_with_predictions
#'    = data where Diabetes_prediction and Correct_prediction
#'    columns (both logical) are added
#' confusion_matrix
#'    = confusion matrix for Actual vs Predicted diabetes
#' correct_proportion
#'    = Proportion of correctly classified observations
#' training_misclass_error
#'    = Proportion of misclassified observations
#' pred_plot
#'    = Plot of Glucose vs Age, colored by Diabetes_predicion
#' class_plot
#'    = Plot of Glucose vs Age, colored by Correct_pred
#'
A3_classifications_and_plots <-
  function(data_pred_prob, r, glucose_range, age_range){

  # adding prediction classifications to the data tibble
  data_with_predictions <-
    data_pred_prob |>
    mutate(Diabetes_prediction = ifelse(Predicted_prob > r, TRUE, FALSE),
           Correct_pred = (Diabetes == Diabetes_prediction))
```

```
# confusion matrix
logistic_confusion <-
  table(Actual = data_with_predictions$Diabetes,
        Predicted = data_with_predictions$Diabetes_prediction)

# proportion correctly classified observations
correct_proportion <-
  sum(diag(logistic_confusion))/sum(logistic_confusion)

# training misclassification error
# = proportion of misclassified observations
training_misclass_error <-
  1 - correct_proportion

# scatterplot colored by predictions
pred_plot <-
  data_with_predictions |>
  ggplot() +
  aes(x = Age, y = Glucose, color = Diabetes_prediction) +
  geom_point(size = 0.3, shape = 8) +
  theme_bw() +
  theme_bw() +
  theme(legend.position = "top") +
  scale_colour_manual(labels = c("Non-diabetic", "Diabetic"),
                      values = c("darkmagenta", "darkorange")) +
  scale_x_continuous(
    limits = age_range + c(-5, +5),
    breaks = scales::breaks_width(10)) +
  scale_y_continuous(
    limits = glucose_range + c(-5, +5),
    breaks = scales::breaks_width(10),
    expand = expansion(mult = c(0, 0.05))
  ) +
  labs(y = "Plasma glucose", color = "Diabetes prediction")

# scatterplot colored by correct or misclassfication
class_plot <-
  data_with_predictions |>
  ggplot() +
  aes(x = Age, y = Glucose, color = Correct_pred, shape = Correct_pred) +
  geom_point(size = 0.3) +
  theme_bw() +
  theme(legend.position = "top") +
  scale_colour_manual(
    name = "Classification result",
    values = c(`TRUE` = "green", `FALSE` = "red"),
    labels = c(`TRUE` = "Correct", `FALSE` = "Misclassification")
```

```
    ) +
    scale_shape_manual(
      name = "Classification result",
      values = c(`TRUE` = 18, `FALSE` = 4),
      labels = c(`TRUE` = "Correct", `FALSE` = "Misclassification")
    ) +
    scale_x_continuous(
      limits = age_range + c(-5, +5),
      breaks = scales::breaks_width(10)) +
    scale_y_continuous(
      limits = glucose_range + c(-5, +5),
      breaks = scales::breaks_width(10),
      expand = expansion(mult = c(0, 0.05))
    )

  return(list(data_with_predictions = data_with_predictions,
              confusion_matrix = logistic_confusion,
              correct_proportion = correct_proportion,
              training_misclass_error = training_misclass_error,
              pred_plot = pred_plot,
              class_plot = class_plot))
}

##############################################################################
#' A3_calculate_linear_boundary
#' Calculating a linear boundary intercept and slope
#'
#' @param theta0 Fitted intercept
#' @param theta1 Fitted coefficient for Glucose
#' @param theta2 Fitted coefficient for Age
#'
#' @returns A list with boundary intercept and slope
#'
A3_calculate_linear_boundary <- function(theta0, theta1, theta2, r){

  # decision boundary line intercept and slope
  boundary_intercept <-  (log(r/(1-r))-theta0)/theta1
  boundary_slope <-  -theta2/theta1

  return(list(boundary_intercept = boundary_intercept,
              boundary_slope =  boundary_slope))
}

##############################################################################
#' A3_add_linear_boundary
#' Adds a decision boundary line to the provided plots
#'
```

```
#' @param results_list List returned by
#' A3_classifications_and_plots
#'
#' @returns A list of two plots,
#' one colored by diabetes prediction, and
#' one colored by the predicion being correct or not
#'
A3_add_linear_boundary <-  function(results_list, r ){
  # calculate intercept and slope for the straight line
  boundary <-
    A3_calculate_linear_boundary(A3_theta0, A3_theta1, A3_theta2, r)

  # add line to prediction plot
  pred_plot_with_line <-
    results_list[["pred_plot"]] +
    geom_abline(
      intercept = boundary[["boundary_intercept"]],
      slope = boundary[["boundary_slope"]],
      color = "black",
      linewidth = 1
    )

  # add line to classification correctness plot
  class_plot_with_line <-
    results_list[["class_plot"]] +
    geom_abline(
      intercept = boundary[["boundary_intercept"]],
      slope = boundary[["boundary_slope"]],
      color = "black",
      linewidth = 1
    )
  return(list(pred_plot_with_line=pred_plot_with_line,
              class_plot_with_line=class_plot_with_line))
}

##############################################################################
```

A.3.2 Assignment 3 — Main script

```
library(tidyverse)

# Data in pima-indians-diabetes.csv:
# 1. Number of times pregnant.
# 2. Plasma glucose concentration at 2 hours in an oral glucose tolerance test.
# 3. Diastolic blood pressure (mm Hg).
# 4. Triceps skinfold thickness (mm).
# 5. 2-Hour serum insulin (mu U/ml).
```

```
# 6. Body mass index (weight in kg/(height in m)^2).
# 7. Diabetes pedigree function.
# 8. Age (years).
# 9. Diabetes (0=no or 1=yes).

# import user-definied functions
source("R/lab1_assignment3_functions.R")

# read data
A3_pimadata <- read.csv("data/pima-indians-diabetes.csv")

# any NA:s?
if (anyNA(A3_pimadata)) {
  message("NA:s in table")
}

# create tibble with meaningful column names
A3_pimatibble_raw <-
  A3_pimadata |>
  tibble() |>
  magrittr::set_colnames(c("Parity", "Glucose", "DBP", "Skinfold",
                           "Insulin","BMI", "Pedigree", "Age", "Diabetes")) |>
  mutate(Diabetes = as.logical(Diabetes)) |>
  mutate(ID = row_number(), .before = Parity) |>
  select(ID, Age, Pedigree, Parity, BMI, Skinfold,
         DBP, Insulin, Glucose, Diabetes)

# data check
  A3_pimatibble_raw |>
  pivot_longer(-c(ID,Diabetes),
               names_to = "Measure",
               values_to = "Value") |>
  ggplot() +
  aes(x=Value, fill = Diabetes) +
  geom_histogram(bins = 30) +
  scale_fill_manual(labels = c("Non-diabetic", "Diabetic"),
                    values = c("royalblue", "maroon2")) +
  facet_wrap(~ Measure, scales = "free", nrow = 2) +
  theme_bw() +
  theme(legend.position = "top")

# checking exact range for Age, Glucose, and Diabetes
A3_pimatibble_raw |>
  select(c(Age, Glucose, Diabetes)) |>
  map(\(x) range(x))

# removing cases with Glucose = 0
```

```
A3_pimatibble <-
  A3_pimatibble_raw |>
  filter(Glucose != 0)

# saving adjusted ranges
A3_minmaxlist <-
  A3_pimatibble |>
  select(Age, Glucose) |>
  map(\(x) tibble(min = min(x), max = max(x)))

A3_Age_range <- unlist(A3_minmaxlist[["Age"]])
A3_Glucose_range <- unlist(A3_minmaxlist[["Glucose"]], use.names = FALSE)

# scatterplot Glucose vs Age

A3_Glucose_vs_Age_scatter <-
  A3_pimatibble |>
  ggplot() +
  aes(x = Age, y = Glucose, color = Diabetes) +
  geom_point(size = 0.5, alpha = 0.8) +
  theme_bw() +
  theme(legend.position = "top") +
  scale_colour_manual(labels = c("Non-diabetic", "Diabetic"),
                      values = c("royalblue", "maroon2")) +
  scale_x_continuous(
    limits = A3_Age_range + c(-5, +5),
    breaks = scales::breaks_width(width=10)) + # equally spaced breaks
  scale_y_continuous(
    limits = A3_Glucose_range + c(-5, +5),
    breaks = scales::breaks_width(10),
    expand = expansion(mult = c(0, 0.05))
  ) +
  labs(y = "Plasma glucose")

# training a logistic regression model

# Diabetes ~ Glucose + Age in a logistic regression model
# using glm()= with family = "binomial" since target has two levels

A3_logistic_model <-
  glm(Diabetes ~ Glucose + Age, data = A3_pimatibble, family = "binomial")

summary(A3_logistic_model)

# coefficients (= coefficients in the probabilistic equation)

A3_logistic_coef <- coef(A3_logistic_model)
```

```r
A3_theta0 <- A3_logistic_coef["(Intercept)"]
A3_theta1 <- A3_logistic_coef["Glucose"]
A3_theta2 <- A3_logistic_coef["Age"]

# adding predicted probabilities to data
A3_pimatibble_pred_probs <-
  A3_pimatibble |>
  mutate(Predicted_prob = predict(A3_logistic_model, type = "response"))

# adding predicted diabetess classification
# and comparison with actual diabetes class
# and plot visualizing the results
A3_results_r_05 <-
  A3_classifications_and_plots(A3_pimatibble_pred_probs,
                               r = 0.5, A3_Glucose_range, A3_Age_range)

A3_results_r_02 <-
  A3_classifications_and_plots(A3_pimatibble_pred_probs,
                               r = 0.2, A3_Glucose_range, A3_Age_range)

A3_results_r_08 <-
  A3_classifications_and_plots(A3_pimatibble_pred_probs,
                            r = 0.8, A3_Glucose_range, A3_Age_range)

# adding decision boundary lines to Glucose vs Age plots
A3_boundary_05_plots <-
  A3_add_linear_boundary(A3_results_r_05, r = 0.5)

A3_boundary_02_plots <-
  A3_add_linear_boundary(A3_results_r_02, r = 0.2)

A3_boundary_08_plots <-
  A3_add_linear_boundary(A3_results_r_08, r = 0.8)

# basis function expansion
# sdding new features
A3_pimatibble_extended <-
  A3_pimatibble |>
  mutate(z1 = Glucose^4,
         z2 = Glucose^3*Age,
         z3 = Glucose^2*Age^2,
         z4 = Glucose*Age^3,
         z5 = Age^4)

A3_logistic_model_extended <-
  glm(Diabetes ~ Glucose + Age + z1 + z2 + z3 + z4 + z5,
      data = A3_pimatibble_extended, family = "binomial")
```

```r
summary(A3_logistic_model_extended)

# coefficients (= coefficients in the probabilistic equation)

A3_logistic_extended_coef <- coef(A3_logistic_model_extended)

A3_theta0_ext <- A3_logistic_extended_coef["(Intercept)"]
A3_theta1_ext <- A3_logistic_extended_coef["Glucose"]
A3_theta2_ext <- A3_logistic_extended_coef["Age"]
A3_theta3_ext <- A3_logistic_extended_coef["z1"]
A3_theta4_ext <- A3_logistic_extended_coef["z2"]
A3_theta5_ext <- A3_logistic_extended_coef["z3"]
A3_theta6_ext <- A3_logistic_extended_coef["z4"]
A3_theta7_ext <- A3_logistic_extended_coef["z5"]


# adding predicted probabilities to data
A3_pimatibble_extended_pred_probs <-
  A3_pimatibble_extended |>
  mutate(Predicted_prob = predict(A3_logistic_model_extended, type = "response"))



A3_results_extended <-
  A3_classifications_and_plots(A3_pimatibble_extended_pred_probs,
                               r = 0.5,
                               A3_Glucose_range, A3_Age_range)
```